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2023 Year 11 Examination

Mathematics Standard

General Instructions

- Reading time – 5 minutes
- Working time – 2 hours
- Write using black pen
- Write your student number on the examination and multiple-choice answer sheet
- Calculators approved by NESA may be used
- A reference sheet is provided at the back of this paper
- For questions in Section II, show relevant mathematical reasoning and/or calculations

Total marks:
80

Section I – 15 marks (pages 2-5)

- Attempt Questions 1–15
- Write your answers on the multiple-choice answer sheet provided
- Allow about 25 minutes for this section

Section II – 65 marks (pages 6-20)

- Attempt Questions 16–39
- Allow about 1 hour and 35 minutes for this section

Question	Section I	Section II	Total
Marks	15	65	80
Student's Score			

Teacher's comment:

Disclaimer

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2023-MS2-NSW-Y11-NA-EX-QATS

Section I
15 marks

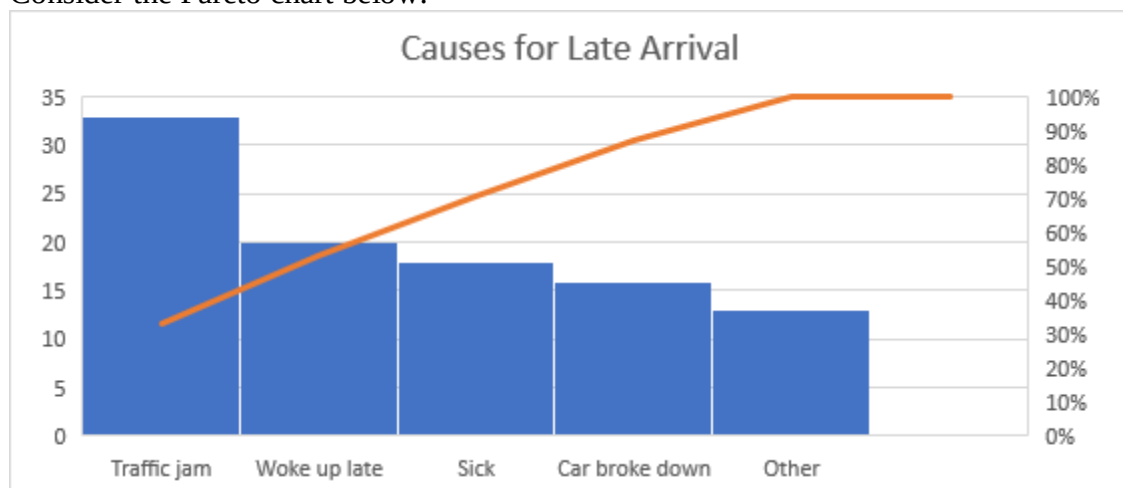
Attempt Questions 1–15
Allow about 25 minutes for this section

Use the multiple-choice answer sheet for Questions 1–15.

- 1** If $w = \frac{15y}{y+12}$ and $y = 11$, find the value of w , correct to 2 decimal places.
- A. 6.89
B. 7.17
C. 8.12
D. 27.01
- 2** The probability of an event occurring is 0.2. What is the probability of the complementary event?
- A. 0.08
B. 20 %
C. 0.2
D. 0.8
- 3** Tyson buys a packet of peanuts weighing 475 grams, measured correctly to the nearest gram. Calculate the percentage error.
- A. 0.5 %
B. 0.11 %
C. 2.5 %
D. 5.0 %
- 4** Express 0.000 005 55 in scientific notation, correct to 2 significant figures.
- A. 5.56×10^{-6}
B. 5.56×10^{-5}
C. 5.56×10^{-3}
D. 5.6×10^{-6}

- 5 Sheryl wishes to call her grandmother for her birthday. Sheryl lives in Perth (UTC +8) and her grandmother lives in Auckland (UTC +12). What time in Perth will Sheryl need to call if she wants it to be 11am in Auckland?
- A. 7 am
B. 3 pm
C. 7 pm
D. 11 pm
- 6 The capacity of a motorcycle's fuel tank is measured as 12.5L, rounded to the nearest 50 mL. What is the absolute error of this measurement?
- A. 10 mL
B. 25 mL
C. 100 mL
D. 250 mL

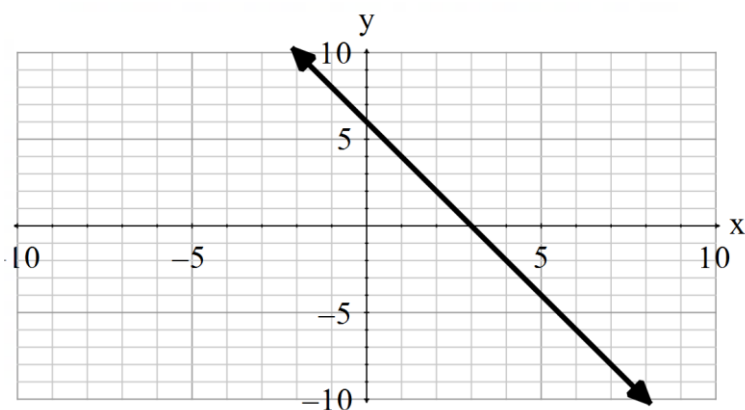
- 7 Consider the Pareto chart below.



What percentage of late arrivals were due to waking up late?

- A. 20 %
B. 31 %
C. 57 %
D. 80 %

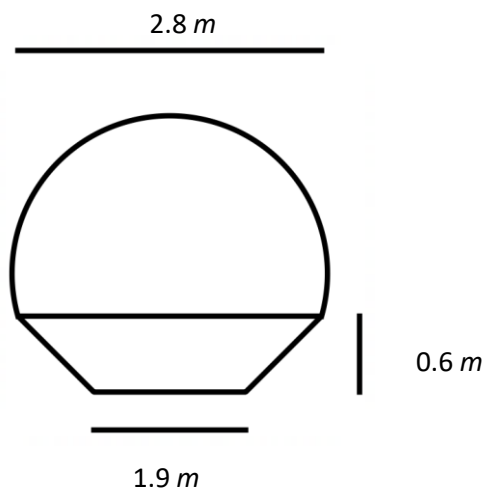
- 8 The five-number summary of a data set is 14, 23, 26, 29 and 40. This data set will have:
- no outlier.
 - at least 1 lower outlier.
 - at least one upper outlier.
 - both upper and lower outliers.
- 9 A clown quoted \$700, excluding GST, to be hired for a carnival. A GST of 10% is added to the price. What is the full price of hiring a clown?
- \$70
 - \$707
 - \$710
 - \$770
- 10 A company depreciates all its vehicles at a rate of 8% p.a. using the straight-line method. A Subaru Impreza was purchased on 1st January 2015 for \$28 500. What was its value on 1st January 2022?
- \$2 280
 - \$12 540
 - \$15 960
 - \$26 220
- 11 What is the equation that describes the graph below?



- $y = -3x + 6$
- $y = -2x + 6$
- $y = 6x + 2$
- $y = 3x + 6$

- 12 Jodi is a photographer who is paid a retainer of \$280 per month, plus a royalty of 2% of the sales of her photographs. In the last six months, these sales totalled \$456 000. What was her average weekly income over the last six months?
- A. \$415
B. \$831
C. \$9 120
D. \$10 800

- 13 A sign for a cupcake is to be made to the dimensions shown below. What is the area of the sign, correct to 2 significant figures?



- A. 1.4 m^2
B. 4.5 m^2
C. 7.6 m^2
D. 10.0 m^2
- 14 Barney's car can be driven for 745 km using 43 litres of fuel. His car has a fuel tank with a capacity of 65 litres. What distance can Barney's car be driven on a full tank of fuel, at the same rate of fuel consumption?
- A. 888 km
B. 910 km
C. 1 126 km
D. 1 885 km
- 15 Which of the following represents continuous numerical data?
- A. The number of soccer teams at a local club
B. The colour of the team's jersey over the past 100 years
C. The total points scored in all the football matches last weekend
D. The time a football team took to score the first goal

End of Section I

Mathematics Standard 2

Section II Answer Booklet

65 marks

Attempt Questions 16–39

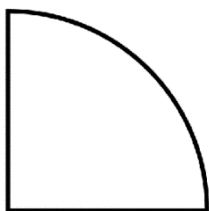
Allow about 1 hour and 35 minutes for this section

Instructions

- Answer the questions in the spaces provided. These spaces provide guidance for the expected length of response.
 - Your responses should include relevant mathematical reasoning and/or calculations.
 - Extra writing space is provided at the back of this booklet. If you use this space, clearly indicate which question you are answering.
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Question 16 (2 marks)

The following is a quarter of a circle with a radius of 7 *cm*. Calculate the area of the sector below, to the nearest square centimetre. 2



NOT TO
SCALE

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Question 17 (2 marks)

A mobile phone plan includes a charge of 97 cents per minute plus 40 cents connection charge for each standard voice call. How much would be charged for 8 calls that were on average 4 minutes long each? 2

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Question 18 (2 marks)

June earns \$2 875 per month. What is her new annual salary if she is given a 3.5 % pay rise? 2

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Question 19 (3 marks)

The probability of Lam getting caught for speeding on a Friday night is 0.4. On the next two Friday nights, Lam is meeting friends and will be driving home.

- (i) What is the probability that he won't be caught for speeding on the first Friday night? 1

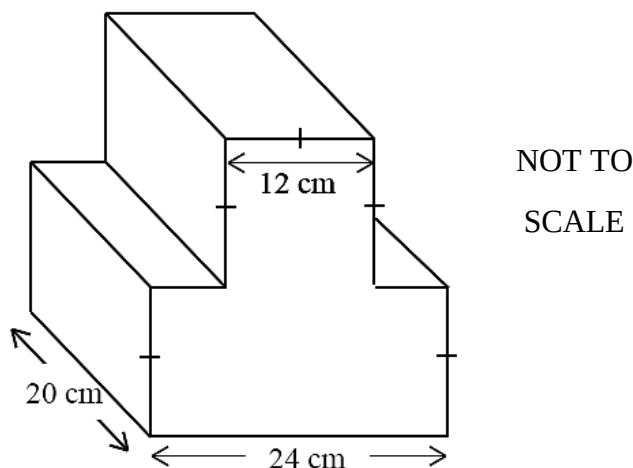
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- (ii) What is the probability that Lam is not caught speeding on one night, however, is caught speeding on the other night? 2

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Question 20 (4 marks)

The diagram shows a prism whose front face consists of a square and a rectangle. The sides of the square have a length of 12 cm. The length of the rectangle is 24 cm and its width is 12 cm. The depth of the prism is 20 cm.



- (i) Find the perimeter of the front face. 1

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Question 20 continues on the next page

Question 20 (continued)

- (ii) Find the area of the front face. 2

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- (iii) Find the volume of the prism. 1

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Question 21 (2 marks)

A stratified random sample survey is to be conducted to determine whether hats should be part of the school uniform. The table shows how many students are in each year group.

Year 7	100
Year 8	120
Year 9	150
Year 10	140
Year 11	110
Year 12	100

- If 50 students are to be interviewed, how many of these should be from Year 12? 2

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Question 22 (2 marks)

Simplify:

$$5(2y - 4) - 3(2y - 5).$$

2

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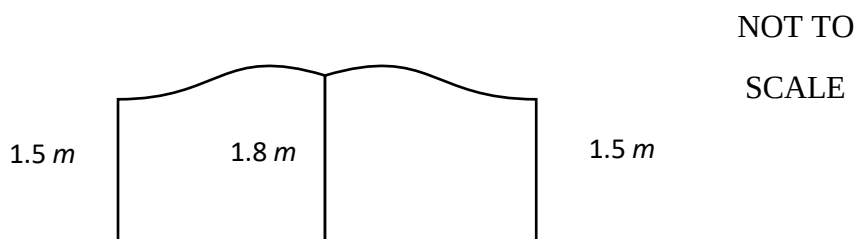
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Question 23 (2 marks)

The paved courtyard shown is split into two sections with a total width of 8 m.
The two sections are of equal width.



Use 2 applications of the trapezoidal rule to approximate the area of the paved area.

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Question 24 (2 marks)

Athens lies on the 24°E line of longitude and Sao Paulo lies on the 47°W line.
Calculate the time difference between the cities, given that 15° = 1 hour, to the nearest minute.

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Question 25 (2 marks)

When a passenger vehicle is sold in NSW, stamp duty is payable when the registration is transferred to the new owner. The rates are given in the table below.

Value of Vehicle (nearest \$100)	Stamp Duty Calculation
\$0 - \$45 000	3% of the value of the vehicle
\$45 100 and over	\$1 350 plus 5% of the vehicle's value over \$45 000

Ian intends to buy a car, which is already registered, in a private sale for \$54 200. How much is the stamp duty on the transfer of registration? 2

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Question 26 (2 marks)

Charlie earns an annual salary of \$55 000. He is given an increase of 2% for one year, and then another 3% for the subsequent year. What is his new salary after 2 years? 2

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Question 27 (2 marks)

Joy asked the students in her class how many brothers they had. The answers were recorded in the frequency table below.

Number of brothers	Frequency
0	5
1	8
2	3
3	2

What is the mean number of brothers?

2

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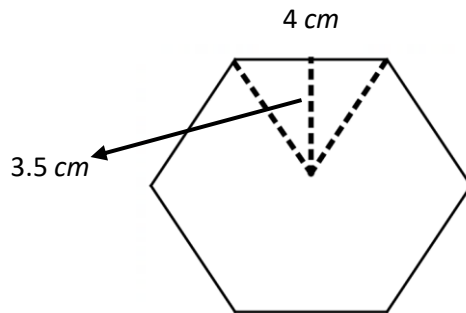
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Go to the next page for Question 28

Question 28 (4 marks)

A hexagon is made up of six identical equilateral triangles with edges of 4 cm and each triangle has a perpendicular height of approximately 3.5 cm.



NOT TO
SCALE

- (i) Calculate the area of the hexagon.

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- (ii) A prism has a hexagon as its cross-section and a height of 25 cm.

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Find the capacity of the hexagonal prism, to the nearest litre.

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Question 29 (2 marks)

Nicky lives in Sydney (UTC +10) and wants to fly to Chicago (UTC -6).
The flight leaves Sydney at 8pm Thursday and has a flight time of 19 hours.
What is the local time and day in Chicago when she lands?

2

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Question 30 (5 marks)

Last year, Anton's taxable income was \$98 750.

Taxable Income	Tax on this Income
0 - \$18 200	Nil
\$18 201 - \$45 000	19 cents for each \$1 over \$18 200
\$45 001 - \$120 000	\$5 092 plus 32.5 cents for each \$1 over \$45 000
\$120 001 - \$180 000	\$29 467 plus 37 cents for each \$1 over \$120 000
\$180 001 and over	\$51 667 plus 45 cents for each \$1 over \$180 000

Source credit: Australian Taxation Office (ATO)

- (i) Calculate his tax payable, using the table above. 2

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- (ii) In addition to tax, Anton must pay the Medicare levy which is set at 2% of his taxable income. 1
How much of the Medicare levy does Anton owe?

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- (iii) During the year, Anton has already paid \$28 080 in PAYG tax instalments to cover his income tax and Medicare levy. 2
After completing his tax return, will Anton receive a tax refund or owe more in tax? Justify your answer with supporting calculations.

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Question 31 (2 marks)

The table below shows the number of kilojoules burned per kilogram of body weight in an hour for various activities.

Activity	kJ/kg/h
Cycling	23.0
Jogging	34.1
Running	64.5
Swimming	22.1
Weight training	35.0

Lily has a body weight of 55 kg and spends two hours on exercise each day. One day she spent forty minutes cycling, ten minutes swimming, and the remaining time jogging. How many kilojoules does she burn in her exercise session?

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Question 32 (2 marks)

Solve the following equation:

$$\frac{x}{2} + 3 = 12.$$

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Question 33 (3 marks)**3**

The time taken (t) in hours for customers to receive their orders at a restaurant is inversely proportional to the square of the number of staff (s) on duty. It takes 20 minutes for customer orders to be taken when there are 4 staff members on duty. If the number of staff is doubled, how long will it take to complete the orders?

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Question 34 (4 marks)

The formula for converting degrees Fahrenheit to degrees Celsius is

$$C = \frac{5}{9}(F - 32).$$

- (i) What is 75 degrees Fahrenheit in degrees Celsius, to the nearest degree?

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- (ii) Rearrange the equation with F as the subject.

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- (iii) What is 40 degrees Celsius in degrees Fahrenheit?

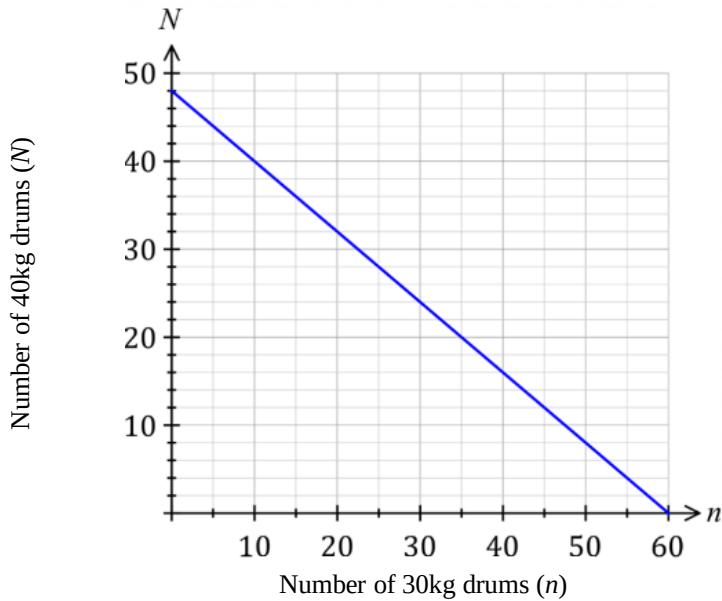
1

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Question 35 (6 marks)

- (a) A truck is transporting drums full of waste weighing either 30 kg or 40 kg. The graph relating the number of 40 kg drums on the truck (N) and the number of 30 kg drums on the truck (n) is shown below.



- (i) What is the total weight of the drums on the truck if $N = 40$? **1**

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- (ii) What is the equation of the line graph? **2**

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Question 35 continues on the next page

Question 35 (continued)

(iii) What is the value of N at point when $n = 20$? **1**

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(iv) What does the gradient of the line and the y -intercept represent? **2**

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Go to the next page for Question 36

Question 36 (3 marks)

Jack and Dianne are charged a peak rate of 26.334 cents per kWh for their electricity and an off-peak rate of 12.199 cents per kWh. Assuming they use their 1200 W dishwasher for half an hour each day, how much will they save annually if they always use it in the off-peak period?

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Question 37 (3 marks)

A loan of \$12 000 is taken out at a simple interest rate of 14.5% per annum. How much is owed, in total, after 3 months?

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Question 38 (2 marks)

Blood alcohol can be estimated using the BAC formula shown below.

$$BAC_{Female} = \frac{10N - 7.5H}{5.5M}$$

If Caitlin's BAC is 0.05 and her weight is 55 kg, over how many hours did she drink 6 standard drinks? **2**

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Question 39 (2 marks)

A jar contains 15 jellybeans in total. Five are yellow, two are blue, four are green and the remaining are red. A jellybean is taken from the jar and not replaced. A second jellybean is taken from the jar, and it is not replaced. What is the probability that both the jellybeans are red? **2**

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End of Section II

End of Examination

Student Number.....

MATHEMATICS STANDARD 2 – MULTIPLE-CHOICE ANSWER SHEET

ATTEMPT ALL QUESTIONS

PLACE A CROSS (X) IN THE BOX TO INDICATE YOUR ANSWER

Question				
1	A	B	C	D
2	A	B	C	D
3	A	B	C	D
4	A	B	C	D
5	A	B	C	D
6	A	B	C	D
7	A	B	C	D
8	A	B	C	D
9	A	B	C	D
10	A	B	C	D
11	A	B	C	D
12	A	B	C	D
13	A	B	C	D
14	A	B	C	D
15	A	B	C	D

Section II extra writing space

If you use this space, clearly indicate which question you are answering.

Mathematics Standard Reference Sheet

Measurement

Limits of accuracy

$$\text{Absolute error} = \frac{1}{2} \times \text{precision}$$

$$\text{Upper bound} = \text{measurement} + \text{absolute error}$$

$$\text{Lower bound} = \text{measurement} - \text{absolute error}$$

Length

$$l = \frac{\theta}{360} \times 2\pi r$$

Area

$$A = \frac{\theta}{360} \times \pi r^2$$

$$A = \frac{h}{2} (a + b)$$

$$A = \frac{h}{2} (d_f + d_l)$$

Surface area

$$A = 2\pi r^2 + 2\pi rh$$

$$A = 4\pi r^2$$

Volume

$$V = \frac{1}{3}Ah$$

$$V = \frac{4}{3}\pi r^3$$

Trigonometry

$$A = \frac{1}{2}ab \sin C$$

$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$$

$$c^2 = a^2 + b^2 - 2ab \cos C$$

$$\cos C = \frac{a^2 + b^2 - c^2}{2ab}$$

Financial Mathematics

$$FV = PV(1 + r)^n$$

Straight-line method of depreciation

$$S = V_0 - Dn$$

Declining-balance method of depreciation

$$S = V_0(1 - r)^n$$

Statistical Analysis

An outlier is a score

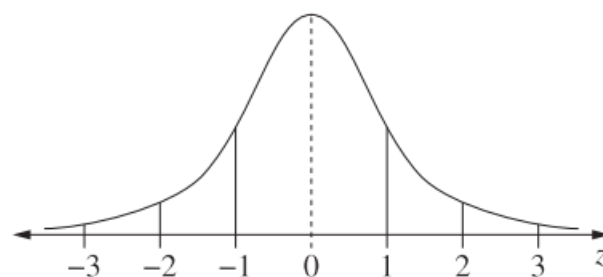
$$\text{less than } Q_1 - 1.5 \times IQR$$

or

$$\text{more than } Q_3 + 1.5 \times IQR$$

$$z = \frac{x - \mu}{s}$$

Normal distribution



- approximately 68% of scores have z-scores between -1 and 1
- approximately 95% of scores have z-scores between -2 and 2
- approximately 99.7% of scores have z-scores between -3 and 3